

MASON COUNTY AP, MI, USA

WMO: 726364

Lat: 43.963N

Lon: 86.408W

Elev: 195

StdP: 99.00

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04883

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.9	-19.9	0.7	-14.2	-17.6	0.8	-12.2	11.6	0.3	10.7	0.6	2.8	310	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	28.6	22.1	27.4	21.2	26.6	20.5	23.6	27.1	22.8	26.0	21.9	25.4	4.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.6	26.1	21.6	16.7	25.3	20.9	15.9	24.6	71.3	27.1	67.8	26.3	65.1	25.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	8.9	8.0	-19.8	31.4	2.9	1.2	-21.9	32.3	-23.6	33.0	-25.2	33.7	-27.4	34.6
			-20.2	25.3	2.8	1.2	-22.2	26.2	-23.8	26.9	-25.4	27.6	-27.5	28.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-4.1	-4.0	0.4	6.5	12.4	17.6	20.1	19.5	16.0	9.6	3.9	-1.0	(6)
	DBStd	9.77	5.15	4.95	5.83	5.02	4.88	3.71	3.23	2.97	4.17	4.68	4.37	4.50	(7)
	HDD10.0	1892	436	392	304	130	29	1	0	0	3	67	191	340	(8)
	HDD18.3	3922	694	625	557	356	193	57	19	22	93	274	434	598	(9)
	CDD10.0	1212	0	0	6	24	104	227	312	293	184	54	7	1	(10)
	CDD18.3	201	0	0	1	1	10	33	73	57	24	3	0	0	(11)
	CDH23.3	1504	0	0	2	6	92	283	556	387	162	16	0	0	(12)
	CDH26.7	206	0	0	0	0	14	44	89	42	16	1	0	0	(13)
(14) Wind	WSAvg	3.3	3.8	3.7	3.5	3.7	3.3	2.8	2.6	2.5	2.7	3.3	3.8	3.8	(14)
(15) Precipitation	PrecAvg	848	51	37	52	74	66	77	54	101	103	86	80	68	(15)
	PrecMax	1075	115	91	180	118	139	211	108	238	272	196	186	163	(16)
	PrecMin	616	11	6	8	25	10	5	16	8	6	26	16	23	(17)
	PrecStd	118	29	23	39	26	35	45	27	57	62	38	39	35	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	11.4	21.1	24.0	28.0	30.1	31.9	29.1	28.7	25.9	18.0	13.5	(19)
		MCWB	8.7	8.4	15.8	15.6	20.2	23.0	25.6	22.6	22.2	20.5	13.6	12.0	(20)
	2%	DB	6.3	7.2	15.6	21.2	26.1	27.7	28.7	27.7	27.1	22.6	15.0	9.0	(21)
		MCWB	4.0	5.3	12.1	13.5	19.1	20.6	22.7	21.9	20.5	18.2	11.7	7.9	(22)
	5%	DB	4.1	4.1	12.2	17.7	23.7	26.6	27.4	27.0	25.2	20.3	12.8	6.6	(23)
		MCWB	2.7	2.2	8.7	11.3	17.7	19.8	21.6	21.2	20.1	16.6	10.0	4.8	(24)
	10%	DB	2.8	2.6	8.8	15.1	21.2	24.7	26.3	25.5	23.8	17.3	11.2	4.9	(25)
		MCWB	1.6	1.1	6.1	9.5	15.4	19.0	20.7	20.4	19.3	14.0	8.6	3.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	8.5	16.2	17.8	22.2	23.9	25.6	24.3	23.3	21.5	14.7	11.7	(27)
		MCDB	9.4	10.4	21.2	22.9	27.2	28.6	30.2	27.9	26.4	25.0	17.3	13.2	(28)
	2%	WB	4.5	5.2	12.4	14.6	20.2	22.1	23.8	23.3	22.0	19.2	12.2	8.0	(29)
		MCDB	5.8	7.2	14.7	19.2	24.7	26.4	27.6	26.3	25.2	21.8	14.2	8.8	(30)
	5%	WB	3.0	2.8	8.8	12.2	18.6	21.0	22.9	22.5	21.0	16.8	10.4	5.2	(31)
		MCDB	4.3	4.1	11.4	16.4	22.6	24.6	26.1	25.5	23.6	19.7	12.6	6.5	(32)
	10%	WB	1.6	1.2	5.8	10.1	16.5	20.0	22.0	21.6	20.0	14.4	8.8	3.4	(33)
		MCDB	2.9	2.4	8.5	14.5	20.0	23.8	25.5	24.6	23.3	17.1	11.0	4.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	8.5	9.9	11.0	10.7	10.1	10.3	10.6	8.9	6.9	5.1	(35)
		MCDBR	7.0	8.3	11.4	14.0	12.8	12.3	10.5	10.8	11.5	11.6	9.8	7.8	(36)
	5% WB	MCWBR	6.3	7.0	8.5	8.5	7.4	6.9	6.0	6.4	6.9	8.1	7.4	7.2	(37)
		MCWBR	7.0	7.8	10.6	12.2	11.3	10.1	8.9	8.7	9.2	10.6	9.0	7.8	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.293	0.312	0.334	0.365	0.396	0.408	0.409	0.401	0.383	0.348	0.316	0.289	(40)
		taud	2.436	2.370	2.395	2.336	2.317	2.325	2.329	2.351	2.384	2.469	2.498	2.485	(41)
	Ebn at Noon	Ebn at Noon	838	880	904	898	876	864	858	853	840	829	805	811	(42)
		Edh at Noon	73	93	104	119	125	125	123	117	104	84	69	64	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.11	1.98	3.53	4.73	5.70	6.26	6.37	5.40	4.23	2.56	1.43	0.88	(44)
	RadStd	RadStd	0.13	0.23	0.37	0.44	0.49	0.42	0.29	0.34	0.36	0.30	0.21	0.12	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page